DATA ITEM DESCRIPTION

Title: ACCEPTANCE TEST PLAN

Number: DI-NUSU-82339
AMSC Number: 10234
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20210331
Limitation: N/A
GIDEP Applicable: No
Project Number: NUSU-2021-001

Use/Relationship:

The Acceptance Test Plan applies to both parts of military standard MIL-STD-188-125, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Ground-Based C4I Facilities Performing Critical, Time-Urgent Missions*. This DID applies to fixed facilities in accordance with Part I (MIL-STD-188-125-1A) and to transportable systems in accordance with Part II (MIL-STD-188-125-2). This DID contains the format, content, and intended-use information for the data deliverable resulting from the acceptance-test tasks described in section 4.4.2 and Appendices A and B of each part of MIL-STD-188-125, as applicable.

The purpose of acceptance testing the HEMP-protection subsystem (HPS) is to ensure that specified HEMP performance characteristics have been met. HEMP acceptance tests, conducted near the conclusion of a hardening construction or installation contract, are tests for the purpose of demonstrating that at least minimum performance requirements of the HPS have been achieved before the subsystem is accepted by the Government from the contractor.

The Acceptance Test Plan describes the test-execution, data-recording, analysis, inspection, and reporting procedures to be used by the contractor to assess and document the results of the HPS' performance. The plan provides the means for the Government to understand and duplicate methods used by the contractor to demonstrate the HPS' performance with respect to MIL-STD-188-125's acceptance-test requirements. The Department of Defense's HEMP-certification authority requires that a documented acceptance test plan be submitted as part of a HEMP-certification request package for a facility or system.

If a portion of the HPS' components has already been acceptance tested by a previous test program, then documentation of those previous tests, including plans, reports, and test data, is attached as an appendix.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The Acceptance Test Plan's classification shall be in accordance with any relevant facility- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Dr. William Linzey, PhD, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

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2. Format. Contractor format is acceptable.

3. Content. The Acceptance Test Plan shall document the test-execution, data-recording, analysis, inspection, and reporting procedures to be used to comply with MIL-STD-188-125's section 4.4.2 and Appendices A and B, including:

a. Shielding effectiveness (SE) test procedures per Appendix A

b. Pulsed current injection (PCI) test procedures per Appendix B

3.1 Administrative data. The Acceptance Test Plan shall contain an administrative section covering the following:

a. Contract number, CDRL number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the test plan

3.2 Summary information. The Acceptance Test Plan shall summarize the following:

3.2.1 Introduction, to include:

a. General objectives of the acceptance test

b. A general description of the HPS' topology to be tested

c. A statement of any assumptions or limitations associated with the planned acceptance testing

d. A summary of the acceptance test's pass and fail requirements per MIL-STD-188-125

3.2.2 Scope. A general description of the overall test-requirements matrix that is used to demonstrate the HPS' compliance with MIL-STD-188-125's performance requirements, including the relative roles of tests, inspections, and analyses.

3.2.3 Methods of demonstration. A general description of the procedures that are used to demonstrate the HPS' compliance with each acceptance test performance requirement as listed in section 3.3 below.

3.2.4 Engineering factors. Any important engineering factors affecting the acceptance-test procedures, such as: physical-accessibility constraints of the facility's or system's topology, resource limitations, safety and security protocols, and precedent reports.

3.3 Detailed information. The overall Acceptance Test Plan shall describe in detail the test procedures and the associated test equipment. The Acceptance Test Plan shall consist of two portions, one each for the SE and PCI testing (as detailed in Appendices A and B of both Part I and Part II of MIL-STD-188-125). These test-plan portions shall be detailed in two separate sections of the overall Acceptance Test Plan.

3.3.1 The overall test plan. The main portion of the test plan shall contain the following detailed information:

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a. An identification of the facility or system to be tested

b. Identification of the proposed test-team organizations and personnel and of the organizations and witnesses that will likely be present

c. A description of the HPS' topology, to include: drawings showing the locations of all points of entry (POEs) and POE protective devices (PPDs) of each low-risk electromagnetic (EM) barrier, equipment layouts inside each low-risk EM barrier, cable interconnections external to each low-risk EM barrier, and a description of each HEMP-protection measure

d. Descriptions of the data acquisition equipment, to include: the manufacturer, model, and serial numbers of each equipment item; detailed calibration procedures for the equipment; and calibration-traceability documentation

e. Descriptions of the test procedures, including: system, subsystem, and circuit configuration requirements; test configuration diagrams; test point locations and descriptions; and step-by-step test procedures

f. Descriptions of methods for measuring noise (instrumental and local or ambient)

g. Identification of any deviations from the requirements of MIL-STD-188-125 that are anticipated, along with an explanation of why these deviations may be necessary

h. Descriptions of the data-management procedures, to include: data quality control procedures, data acceptability criteria, annotation and preservation of data records, and the methods to compare results to pass and fail criteria

i. Identification of potential safety risks and descriptions of procedures to mitigate these risks, including electromagnetic radiation hazards and electrical shock hazards

j. Descriptions of the data-security and physical-security procedures that are applied

k. Description of the planned test schedule, including a listing of the prioritization of measurements

3.3.2 The SE portion of the test plan. This portion of the Acceptance Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix A's section A.4.7

b. Identification, with diagrams, of the proposed locations and orientations of the facility or system under test with respect to each transmit antenna's and receive antenna's position and polarization

c. A table listing all planned test point locations

d. Identification of the expected measurement-range sensitivities of the receive antennas and data acquisition system

e. Identification of the equations to be used to verify the HPS' SE performance

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3.3.3 The PCI portion of the test plan. This portion of the Acceptance Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with MIL-STD-188-125 Appendix B's section B.4.7 and Tables B-II and B-III

b. Identification, with diagrams, of the planned locations of each HPS subsystem component to be subjected to PCI injection and an explanation for why these specific points were selected

c. Identification of the expected measurement-range sensitivities of the current-measurement probes and data acquisition system

d. A table listing the associated PCI drive levels per test point

e. All PCI testing that is to be performed in the laboratory as opposed to being tested at the facility or system (e.g., the acceptance testing of power filters in the laboratory) shall be noted

3.4 Other information sources. When other available information sources contain data required by the Acceptance Test Plan, these sources shall be referenced, rather than duplicated, within the test plan.

End of DI-NUSU-82339

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